



AADITYA KUMAWAT

Third Year Undergraduate, IIT Delhi
Department of Mechanical Engineering
B.Tech. Production and Industrial Engineering

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Examination	Degree/Certificate	Institute	Year	CGPA/%
Graduation	B.Tech., PIE	Indian Institute of Technology Delhi	2024-Present	7.23/10
Intermediate/+2	CBSE (XII)	Tagore International School, Jaipur	2024	86.8%
Matriculation	RBSE (X)	Nalanda Public School, Jaipur	2022	91.0%

SCHOLASTIC ACHIEVEMENTS

- **Competitive Programming:** Reached **Candidate Master** on **Codeforces** at a peak rating of **1938** (@codeleon) in **9 rated contests** over six weeks, climbing from unrated; best finish **rank 170** in Educational Round 193 [2026]
- Secured an All India Rank in the **top 3%** of the **~200,000** candidates who qualified **JEE Advanced 2024** [2024]
- Scored a **99.33 percentile** in **JEE Main 2024**, placing in the **top 0.7%** nationally out of **~1,500,000** candidates [2024]
- **Letter of Recommendation** from the **Applied AI Laboratory, HEC Lausanne** [2026]

RESEARCH EXPERIENCE

HEC Lausanne, Switzerland (May - Jul 2026)
Research Intern, Applied AI Laboratory • Legal AI

- Engineered a **fully local, API-free** legal-audio pipeline (**Whisper** large-v3-turbo ASR → **pyanote** diarization → 4-bit quantized LLM) producing speaker-attributed, **citation-backed** notes on a consumer 8GB GPU, with a run classifier that **refuses any non-loopback endpoint** so privileged audio never leaves the machine.
- Designed a **model-selected extractive** mode in which the LLM emits **only integer utterance ids** under a decode-time grammar constraint and note text is re-materialised from the transcript, making fabrication **structurally impossible** at a verified **100% verbatim** rate.
- Built the evaluation harness (WER, DER, **Check-Eval** judge, injection tests) over 62 minutes of gold Oyez **Supreme Court** audio, measuring **6.9% WER** and **7.9% DER**, and showing legally load-bearing tokens are **3.0×** more likely to be substituted than the average token.
- Added a **speaker-label bias probe** with a resampling noise floor which found note selection shifting one speaker's share by **40.4 points** under relabelling alone - a failure invisible to WER, DER and verbatim rate.

TECHNICAL SKILLS

- **Languages:** C++20, Python, TypeScript
- **ML:** PyTorch, Hugging Face Transformers, scikit-learn, LightGBM, XGBoost, Optuna, NumPy, pandas, Matplotlib
- **GPU & Inference:** CUDA, Triton, INT4/INT8 quantization
- **Web & Backend:** Next.js, React, PostgreSQL
- **Tools:** Git, Docker, CMake, ASan/UBSan/TSan

RELEVANT COURSES

- Introduction to Computer Science (COL100) • Calculus (MTL100) • Linear Algebra & Differential Equations (MTL101) • Numerical Methods & Computation (MTL107, *ongoing*) • Probability & Statistics (MTL108)

PROJECTS

Sextant - Ontology & Lineage Platform

(Jun 2026 - Aug 2026)

Self-led Project • C++20, React, TypeScript, CMake

- Built an **ontology, entity-resolution and cell-level lineage** platform on a **from-scratch LSM storage engine** (WAL, memtable, SSTables, bloom filters, leveled compaction) sustaining **1.53M batched writes/sec** at **1.20×** write amplification; **418 tests** green on Linux, Windows and macOS, clean under **ASan, UBSan and TSan**.
- Made lineage an **invariant, not a comment**: all **4,201 properties** across **1,187 entities** replay from their raw source row and transform chain - **100% verified**, against a negative control.
- Resolved **1,451** heterogeneous records into **1,187** entities at **F1 0.991** with **veto-constrained clustering** (precision **0.973** → **1.000**), and wrote a **cost-accounting query planner** answering a 3-month window in **359 μs** over 25 keys where a full scan walks 92.

nanoserve - LLM Inference Server

(Jun 2026 - Jul 2026)

Self-led Project • PyTorch, Triton, CUDA

- Built an LLM inference server **from scratch with no vLLM/TGI**: a **paged KV cache** with block allocator and block tables, **continuous batching** with chunked prefill and preemption-by-recompute, and weight-only **INT4/INT8** quantization, covered by **92 tests**.
- Wrote a **fused Triton paged-attention kernel** using online softmax over the block table, measuring **3.4-12.5×** over gather + SDPA with **33% fewer kernel launches** per decode step.
- Delivered **1.65×** throughput and **12×** lower **p99 TTFT** (60.8s → 5.2s) over static batching, cross-validated against **vLLM**; a **roofline probe** showed decode here is **~80% launch-overhead bound**, not bandwidth bound.

QRT Quant Challenge - Return Prediction

(Mar 2026)

Self-led Project • LightGBM, XGBoost, Optuna

- Built a **leakage-safe** pipeline forecasting next-day equity return direction across **143 features** and **51k rows**, using **5-fold purged CV** with a **20-day embargo** and a 72-day untouched holdout; an Optuna-tuned **LightGBM/XGBoost** blend reached **54.1% accuracy** at **IC +0.109**.
- Benchmarked against a simulator **oracle** that knows the true alpha, capturing **73.7% of attainable IC**, and proved the pipeline honest with two controls: **pure-noise data scores at chance**, and dropping the embargo **inflates accuracy by 0.8 points** - a third of the real edge.

ST-BEMD - Direction-Adaptive 2-D EMD

(May 2026 - Jul 2026)

Self-led Project • Python, NumPy, SciPy

- Devised **ST-BEMD**, a **structure-tensor-guided** decomposition building a **locally oriented elliptical envelope**, benchmarked against **4 EMD baselines** written from scratch: mean orientation error cut **35.7%** (12.87° → 8.27°) over **100 SOCOFing fingerprints** (**Wilcoxon p = 1.9 × 10⁻⁹**) at **3.2×** the speed.
- Ran an **ablation isolating the two changes**, attributing **78%** of the gain to the envelope over the anisotropy the method is named for.

Company Hub - Interview Prep Platform

(Jun 2026 - Present)

Self-led Project • Live at companyhub.fun • Next.js 15, TypeScript, PostgreSQL

- Shipped and now operate a **live interview-prep** platform indexing **3,400 questions** across **656 companies**, on a **21-model Prisma/PostgreSQL** schema: **Auth.js v5** (Google OAuth, **TOTP 2FA**), **LeetCode sync**, graded spaced repetition and a **Monaco** workspace across **9 languages**.
- **Hardened it under real production load**: eliminated PostgreSQL **connection-pool exhaustion** on serverless and **P2003** foreign-key violations from stale JWTs; added **rate limiting** and lockout.

Survey: Towards Reliable LLM-as-a-Judge Systems

(2026)

Research Paper • Prof. Amartansh Dubey • LLM Evaluation, AI Security

- Surveyed **five directions** in LLM-based evaluation - **MCTS** reasoning judges, **pairwise vs. pointwise** protocols, **checklist** assessment, **multilingual** judges and **prompt-injection** security - synthesising them into one framework covering evaluation bias and adversarial robustness.